

Beyond the Heatmap: Toward Metaphorical Visualization of Embodied Carbon in Architectural Design

Systematic Literature Review Process

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This document outlines the Systematic Literature Review (SLR) methodology proposed in the paper “Beyond the Heatmap: Toward Metaphorical Visualization of Embodied Carbon in Architectural Design”. This corresponds to Phase 1 of the research, which focuses on surveying conventional building data visualization techniques against metaphorical strategies.

Protocol

To structure the SLR using the PICOC framework, we first defined the PICOC Elements, Research Questions, and Digital Databases utilized for the review. Subsequently, Inclusion/Exclusion Criteria and Quality Assessment protocols were established. Keywords were selected to perform systematic bibliographical searches within the chosen digital libraries. Articles retrieved through this search underwent an initial Screening Process based on the defined criteria and Quality Assessment. The included ones were then subject to a full-text eligibility review.

1. PICOC Elements

The overall research objective was structured using the PICOC framework—Population, Intervention, Comparison, Outcomes, and Context—to facilitate the development of research questions and to identify relevant keywords for the search strategy.

PICOC Elements

POPULATION	Non-experts
INTERVENTION	LCA-data Representation
COMPARISON	Visualization Methods
OUTCOME	LCA Visual Metaphors Design Comprehension Tangible LCA Results
CONTEXT	Early-Stage Decision-Making Architecture Decision-Making

2. Research Questions

Three research questions were formulated to align with the elements of PICOC framework. These questions guided the entire research structure and served as the focal points for the full-text review and analysis of the selected articles.

Research Questions

RQ1. What class and type of limitations are documented regarding data visualization, and what principles are explored to address those limitations?

RQ2. What existing precedents of metaphorical or non-abstract data representation exist in architecture, graphic design, or other discipline found can inform and substantiate possible visual directions for LCA representation?

RQ3. Which graphic limitations and principles were discussed in terms of altering the intended target audience that has access to the communicated data?

3. Digital Database

Digital Libraries

DATABASE	URL
Scopus	https://www.scopus.com/
Scielo	https://www.scielo.org/pt-br/
Redalyc	https://www.redalyc.org/

4. Inclusion / Exclusion Criteria

The Inclusion / Exclusion Criteria served as the initial filtering mechanism applied prior to the abstract review phase.

Inclusion - Exclusion Criteria

CRITERIA TYPE	DEFINITION
Period	Inclusion: From 2016 to 2026 Exclusion: Prior to 2016
Language	Inclusion: English, Portuguese, Spanish Exclusion: Other
Type of Literature	Exclusion: Grey literature - Reports, Policy Literature, Web articles, Working papers, Newsletters, Reviews
Type of Source	Inclusion: Articles from Conferences or Journals
Accessibility	Exclusion: Not accessible
Relevance to research questions	Exclusion: Not relevant to at least 2 research questions

5. Quality Assessment

The Quality Assessment was structured as a secondary screening mechanism, conducted following the abstract review phase. To be selected for full-text review, articles were required to achieve a high agreement score (3–4) across at least three of the evaluation factors. Consequently, those articles meeting this threshold proceeded to the full-text eligibility stage.

Quality Assessment

Relevance	Does it have a strict relationship with the fields of architecture or graphic design? [consider the context and population]	Level of Agreement - Strongly disagree (0) - Disagree (1) - Neither agree nor disagree (2) - Agree (3) - Strongly agree (4)
Rigor	Does it have a clear goal statement based on understanding/facilitating/improving the impact of visualization beyond conventional ones? [consider intervention]	
Credibility	Is the method applied tailored to assess specific aspects of visual representation? [consider comparison]	
Reporting	Are there conclusions concerning the characteristics of the visual representations considered by the paper context? [consider the outcomes]	

6. keywords

Two tracks were employed across Scielo, Redalyc, and Scopus: a "General Approach" to catalog multidisciplinary metaphorical representations, and a "Specific Approach" to examine architectural applications. For each track, a specific group of strings was determined.

Search Strings

General Approach	("visual" OR "graphical") AND "metaphor**" AND "Data Visualization" OR "Data Art" OR "Information Visualization" OR "metaphor* visualization"
Specific Approach	("Decision Making" OR "Decision Support Systems") AND Visualization AND Architecture OR Building OR "architectural design" OR "built environment" OR BIM AND "Data Visualization" OR "Data Art" OR "Information Visualization"

7. Database Search

Upon completion of the preceding stages, the keywords for each approach were searched within the designated digital databases, yielding the results presented in the tables below. The first table lists the articles identified through the 'General Approach,' while the second details those retrieved from the 'Specific Approach.' A total of 100 articles were identified, with 56 resulting from the General Approach and 44 from the Specific Approach. To facilitate the review process, each article was assigned a unique identifier (1–100). The tables incorporate a color-coding system for these IDs to denote the status and classification of each article, as detailed in the legend below.

	Articles selected for full-text review
	Articles excluded based on inclusion/exclusion criteria
	Articles that passed the initial screening but were not included for the full-text review

General Approach

	Article	URL
01	Evaluating Visualization Techniques for Binary Populations in High-Dimensional Evolutionary Search	https://www.scopus.com/pages/publications/105032757914?origin=resultslist
02	Heart Garden: Visualizing Heart Rate Data Using the Metaphor of a Garden	https://www.scopus.com/pages/publications/105007007985?origin=resultslist
03	DATA VISUALIZATION AS ARTISTIC EXPRESSION	https://www.scopus.com/pages/publications/105027334185?origin=resultslist
04	Nodes, Edges, and Artistic Wedges: A Survey on Network Visualization in Art History	https://www.scopus.com/pages/publications/105006592023?origin=resultslist
05	Fluidly Revealing Information: A Survey of Un/foldable Data Visualizations	https://www.scopus.com/pages/publications/105006450974?origin=resultslist
06	ClayVolume: A progressive refinement interaction system for immersive visualization	https://www.scopus.com/pages/publications/105000155956?origin=resultslist
07	Organizing Microbial Diversity and Interspecies Relations through Diagrams: Trees, Maps, and the Visual Semiotics of the Living	https://www.scopus.com/pages/publications/85204430454?origin=resultslist
08	Improve Usability of Multidimensional Data Exploration with Water Fountain Based 3D User Interface Metaphor	https://www.scopus.com/pages/publications/85206256564?origin=resultslist
09	Designing complex network visualisations using the wayfinding map metaphor	https://www.scopus.com/pages/publications/85202211165?origin=resultslist
10	Visually encoding orthogonal planar graph drawings as graph mazes: An eye tracking study	https://www.scopus.com/pages/publications/85204429181?origin=resultslist
11	Pacemod: parametric contour-based modifications for	https://www.scopus.com/pages/publications/85

	glyph generation	167778695?origin=resultslist
12	Visualization Metaphors in the Tasks of Exploratory Analysis of Heterogeneous Data	https://www.scopus.com/pages/publications/85213067064?origin=resultslist
13	Wizualization: A 'Hard Magic' Visualization System for Immersive and Ubiquitous Analytics	https://www.scopus.com/pages/publications/85176325175?origin=resultslist
14	Visualization in Data Reconstruction Tasks	https://www.scopus.com/pages/publications/85191156144?origin=resultslist
15	Visual analysis of inscriptions in the Tang Dynasty: a case study on the calligraphy style of Wang Xizhi	https://www.scopus.com/pages/publications/85196012327?origin=resultslist
16	Visualizing ordered bivariate data on node-link diagrams	https://www.scopus.com/pages/publications/85171992867?origin=resultslist
17	Interactive Input and Visualization for Planning with Temporal Uncertainty	https://www.scopus.com/pages/publications/85148878223?origin=resultslist
18	The Role of Subjective Perception in Visualization, Controllability Hypothesis and Abstract Visualization	https://www.scopus.com/pages/publications/85138878032?origin=resultslist
19	Real-Time Visual Analysis of High-Volume Social Media Posts	https://www.scopus.com/pages/publications/85118619839?origin=resultslist
20	The Datatext: A Multilevel-discursive Theory For Improved Public Health Data Visualizations	https://www.scopus.com/pages/publications/85131778810?origin=resultslist
21	SplitStreams: A Visual Metaphor for Evolving Hierarchies	https://www.scopus.com/pages/publications/85079694896?origin=resultslist
22	A Generic Framework and Library for Exploration of Small Multiples through Interactive Piling	https://www.scopus.com/pages/publications/85100339480?origin=resultslist
23	The State of the Art in Empirical User Evaluation of Graph Visualizations	https://www.scopus.com/pages/publications/85122696232?origin=resultslist
24	Visualization of Emergency Needs Posted on Social Media by Metaphor Map	https://www.scopus.com/pages/publications/85144413720?origin=resultslist
25	Dendrochronology of U.S. Immigration	https://www.scopus.com/pages/publications/85082737924?origin=resultslist
26	Developing Virtual Reality Visualizations for Unsteady Flow Analysis of Dinosaur Track Formation using Scientific Sketching	https://www.scopus.com/pages/publications/85082737924?origin=resultslist
27	Stroke-based splatting: an efficient multi-resolution point cloud visualization technique	https://www.scopus.com/pages/publications/85024494681?origin=resultslist
28	Multidimensional data exploration by explicitly controlled animation	https://www.scopus.com/pages/publications/85058583973?origin=resultslist
29	STAR: Visual Computing in Materials Science	https://www.scopus.com/pages/publications/85022211630?origin=resultslist
30	MoshViz: A Detail+Overview Approach to Visualize Music Elements	https://www.scopus.com/pages/publications/84993972121?origin=resultslist
31	VideoMap: An interactive and scalable visualization for exploring video content	https://www.scopus.com/pages/publications/85040908553?origin=resultslist
32	Designing a situational awareness information display:	https://www.scopus.com/pages/publications/85

	Adopting an affordance-based framework to amplify user experience in environmental interaction design	071846747?origin=resultslist
33	Materializing data: notes on collaboration and tangible interfaces with excerpts and additions	https://www.redalyc.org/articulo.oa?id=758079701001
34	Exploring the Relevance of Europeana Digital Resources: Preliminary Ideas on Europeana Metadata Quality1	https://www.redalyc.org/journal/1790/179049529006/
35	Actor-network Theory and cartography of controversies in Information Science1	https://www.redalyc.org/journal/3843/384357984010/
36	Analyzing the OpenAI Chatbot writing on Economic Development: The case for High Pyrenees and Aran Region and IDAPA	https://www.redalyc.org/journal/5475/547578432009/
37	Graficación Estadística y Visualización de Datos	https://www.redalyc.org/journal/467/46754522005/
38	Suite de componentes para la visualización de textos	https://www.redalyc.org/journal/3783/378360709013/
39	EVOLUCIÓN Y SEMIÓTICA DE LAS INTERFACES GRÁFICAS DIGITALES	https://www.redalyc.org/journal/4779/477963932007/
40	Transposiciones en la construcción de lo invisible: Atti Fondamentali: Life y Digestible Gulf Stream	https://www.redalyc.org/journal/7737/773782606008/
41	La experimentación en el arte audiovisual y en el cine: El lenguaje narrativo abstracto como experiencia temporal, de ruptura de un modelo espectacular y de masas	https://www.redalyc.org/journal/6958/695876700004/
42	Diálogos entre a arquitetura da informação e os primórdios da visualização de dados	https://www.redalyc.org/articulo.oa?id=758081634011
43	Digitalización, metaverso y la (in)distinción entre lo online y lo offline	https://www.redalyc.org/journal/5315/531572894005/
44	Trayectos en el mapa: artes visuales como representación del conocimiento	https://www.redalyc.org/articulo.oa?id=513554412012
45	Complejidad semántica desde la metáfora en la obra surrealista del artista visual ecuatoriano Beto Val	http://www.scielo.org.ar/scielo.php?script=sci_arttext&pid=S1853-35232025001000021&lang=pt
46	School in black and white Photographic images of education in Italy, between reality and metaphor	http://www.scielo.br/scielo.php?script=sci_arttext&pid=S2238-00942025000100413&lang=pt
47	La integración de la alfabetización mediática e informacional en el currículum de Uruguay: conceptualización y presencia en la educación básica integrada	http://www.scielo.edu.uy/scielo.php?script=sci_arttext&pid=S0797-36912025000101205&lang=pt
48	Mobile knowings: Balancing spatial inequities in South African higher education	http://www.scielo.org.za/scielo.php?script=sci_arttext&pid=S2520-98682025000400009&lang=pt
49	Sivivane sogogo!: An artistic reimagining beyond shadow and stone	http://www.scielo.org.za/scielo.php?script=sci_arttext&pid=S1021-14972025000100014&lang=pt
50	Narrar espaços construídos por constelações de	http://www.scielo.cl/scielo.php?script=sci_arttext

	imagens	t&pid=S0719-54272024000200070&lang=pt
51	Queering the soil: Reimagining landscape and identity in queer artistic practices in Cyprus	http://www.scielo.org.za/scielo.php?script=sci_arttext&pid=S1021-14972022000100020&lang=pt
52	Do imperfeito ao mais-que-perfeito: a kalokagathía dos atletas paralímpicos	http://www.scielo.br/scielo.php?script=sci_arttext&pid=S1982-89182018000301001&lang=pt
53	Eisenstein's visual revolution of the art in Ivan, the Terrible	http://www.scielo.br/scielo.php?script=sci_arttext&pid=S1678-53202018000200147&lang=pt
54	El caleidoscopio en la Literatura Española	http://www.scielo.sa.cr/scielo.php?script=sci_arttext&pid=S2215-26362018000100065&lang=pt
55	Conversas com Jovens e Escolas que Passam pelos Filmes e por Nossas Vidas	http://www.scielo.br/scielo.php?script=sci_arttext&pid=S2175-62362017000200455&lang=pt
56	O álbum poético em português e em espanhol: sinergia estética entre palavras e ilustrações	http://www.scielo.br/scielo.php?script=sci_arttext&pid=S1413-24782017000400203&lang=pt

Specific Approach

	Article	URL
57	A BIM-Centered Multi-Source Image Fusion Framework for Remote Client Site Visits	https://www.scopus.com/pages/publications/105032746920?origin=resultslist
58	Research on real-time data display and production management in a digitalized management factory with an artificial intelligence-assisted flexible manufacturing execution system	https://www.scopus.com/pages/publications/105030057348?origin=resultslist
59	Simultaneous Digital Twin: Chaining Climbing-Robot, Defect Segmentation, and Model Updating for Building Facade Inspection	https://www.scopus.com/pages/publications/105030171344?origin=resultslist
60	https://www.scopus.com/pages/publications/105019962761?origin=resultslist	https://www.scopus.com/pages/publications/105019962761?origin=resultslist
61	Federated Data Modelling for Heritage Building Performance Management	https://www.scopus.com/pages/publications/105027871825?origin=resultslist
62	Meta-Learning Enhanced 3D CNN-LSTM Framework for Predicting Durability of Mechanical Metal–Concrete Interfaces in Building Composite Materials with Limited Historical Data	https://www.scopus.com/pages/publications/105021597278?origin=resultslist
63	BIM-Based Integration and Visualization Management of Construction Risks in Water Pumping Station Projects	https://www.scopus.com/pages/publications/105019227368?origin=resultslist
64	INFRASTRUCTURE MANAGEMENT VIA BIM MODEL: INTEGRATION OF STRUCTURAL HEALTH MONITORING AND ANN-BASED DAMAGE ASSESSMENT	https://www.scopus.com/pages/publications/105026345859?origin=resultslist
65	Residential-IST: A web-based decision-support tool for bridging knowledge gaps in performance-driven	https://www.scopus.com/pages/publications/105015136743?origin=resultslist

	residential building design and retrofitting in Istanbul, Türkiye	
66	Utility-inspired generalizations of TOPSIS	https://www.scopus.com/pages/publications/105004199104?origin=resultslist
67	Visualizing ESG Performance in an Integrated GIS–BIM–IoT Platform for Strategic Urban Planning	https://www.scopus.com/pages/publications/105017117212?origin=resultslist
68	A Framework for Data Lifecycle Model Selection	https://www.scopus.com/pages/publications/105017418910?origin=resultslist
69	A real-time VR-enabled digital twin framework for multi-user interaction in Industry 4.0	https://www.scopus.com/pages/publications/105014935027?origin=resultslist
70	Shaping the future of forestry in Germany: Stakeholder perspectives on optimizing forest management through Augmented Reality	https://www.scopus.com/pages/publications/105004246411?origin=resultslist
71	A Web-Based Application for Smart City Data Analysis and Visualization	https://www.scopus.com/pages/publications/105006413884?origin=resultslist
72	Digital occupancy assessment for lighting evaluation: a pilot study to prepare for real-time research results	https://www.scopus.com/pages/publications/85198386108?origin=resultslist
73	Digital Architectural Form Generation Through Pixel System-Driven Image Feature Information	https://www.scopus.com/pages/publications/85210260232?origin=resultslist
74	The development of a novel decision support system for regional land use planning for brownfield land	https://www.scopus.com/pages/publications/85176213218?origin=resultslist
75	A REVIEW OF DRIVERS AND BARRIERS OF DIGITAL TWIN ADOPTION IN BUILDING PROJECT DEVELOPMENT PROCESSES	https://www.scopus.com/pages/publications/85189373648?origin=resultslist
76	Data Fusion for Smart Civil Infrastructure Management: A Conceptual Digital Twin Framework	https://www.scopus.com/pages/publications/85177616082?origin=resultslist
77	Towards a cyber-physical system for sustainable and smart building: a use case for optimising water consumption on a SmartCampus	https://www.scopus.com/pages/publications/85127403439?origin=resultslist
78	Automated multiclass structural damage detection and quantification using augmented reality	https://www.scopus.com/pages/publications/85162203870?origin=resultslist
79	Digital Twin framework for automated fault source detection and prediction for comfort performance evaluation of existing non-residential Norwegian buildings	https://www.scopus.com/pages/publications/85145262458?origin=resultslist
80	Smart real time wearable navigation support system for BVIP	https://www.scopus.com/pages/publications/85135160613?origin=resultslist
81	MetaExplorer: Collaborative development of urban metabolism platform for decision making support	https://www.scopus.com/pages/publications/85144616034?origin=resultslist
82	Intelligent power grid monitoring and management strategy using 3D model visual computation with deep learning	https://www.scopus.com/pages/publications/85126325066?origin=resultslist
82	Research on management informatization construction of electric power enterprise based on big data technology	https://www.scopus.com/pages/publications/85132935958?origin=resultslist

84	Integrating Human Factors in the Visualisation of Usable Transparency for Dynamic Risk Assessment	https://www.scopus.com/pages/publications/85135983707?origin=resultslist
85	LineageD: An Interactive Visual System for Plant Cell Lineage Assignments based on Correctable Machine Learning	https://www.scopus.com/pages/publications/85136252922?origin=resultslist
86	An end-to-end big data analytics platform for IoT-enabled smart factories: A case study of battery module assembly system for electric vehicles	https://www.scopus.com/pages/publications/85127272797?origin=resultslist
87	BIM-based decision support for building condition assessment	https://www.scopus.com/pages/publications/85121962063?origin=resultslist
88	Decision-Making Tool for the Selection of Priority Areas for Building Rehabilitation in Barcelona	https://www.scopus.com/pages/publications/85125078631?origin=resultslist
89	Civil Engineering Simulation and Safety Detection of High-Rise Buildings Based on BIM	https://www.scopus.com/pages/publications/85135724426?origin=resultslist
90	Cognitive analytics platform with AI solutions for anomaly detection	https://www.scopus.com/pages/publications/85118864171?origin=resultslist
91	A SYSTEMATIC LITERATURE REVIEW OF BUSINESS INTELLIGENCE FRAMEWORK FOR TOURISM ORGANIZATIONS: FUNCTIONS AND ISSUES	https://www.scopus.com/pages/publications/85148325412?origin=resultslist
92	Enhancing occupants' comfort through BIM-based probabilistic approach	https://www.scopus.com/pages/publications/85098730352?origin=resultslist
93	A data-driven energy platform: From energy performance certificates to human-readable knowledge through dynamic high-resolution geospatial maps	https://www.scopus.com/pages/publications/85097865913?origin=resultslist
94	Open Visualization Environment (OVE): A web framework for scalable rendering of data visualizations	https://www.scopus.com/pages/publications/85086998949?origin=resultslist
95	Unsupervised learning for feature projection: Extracting patterns from multidimensional building measurements	https://www.scopus.com/pages/publications/85086724078?origin=resultslist
96	Summit: Scaling Deep Learning Interpretability by Visualizing Activation and Attribution Summarizations	https://www.scopus.com/pages/publications/85075960383?origin=resultslist
97	Visual Analytics in Deep Learning: An Interrogative Survey for the Next Frontiers	https://www.scopus.com/pages/publications/85047999594?origin=resultslist
98	A Big Data platform for smart meter data analytics	https://www.scopus.com/pages/publications/85060310485?origin=resultslist
99	An integrated architecture for future studies in data processing for smart cities	https://www.scopus.com/pages/publications/85016030075?origin=resultslist
100	The Role of Uncertainty, Awareness, and Trust in Visual Analytics	https://www.scopus.com/pages/publications/84946595461?origin=resultslist

8. Screening

During the abstract screening phase, as previously noted, we utilized a scoring system based on the Quality Assessment (QA) protocol. Articles achieving a high agreement score (3–4) across at least three evaluation factors were selected for inclusion in the SLR and proceeded to the full-text review stage.

The table below illustrates the process during this phase; each article is identified by its unique ID, followed by the source digital library, individual QA scores, language, accessibility, and the final inclusion/exclusion status.

Articles with ambiguous Quality Assessment results underwent a secondary evaluation conducted by the lead author to determine their final status.

T _r	Article	Source	Relevance	Reporting	Rigor	Credibility	Language	Accessibility	Status	Second evaluation	Final Status
01		Scopus	0	4	4	4	English	Accessible	Included	Yes	Excluded
02		Scopus	1	2	1	2	English	Accessible	Excluded		Excluded
03		Scopus	1	4	3	4	English	Accessible	Included		Included
04		Scopus	0	2	2	2	English	Not Accessible	Excluded		Excluded
05		Scopus	0	3	4	4	English	Accessible	Included		Included
06		Scopus	0	3	1	3	English	Accessible	Excluded	Yes	Excluded
07		Scopus	0	3	0	2	English	Accessible	Excluded		Excluded
08		Scopus	2	4	4	4	English	Accessible	Included		Included
09		Scopus	1	3	3	3	English	Accessible	Included		Included
10		Scopus	0	3	3	3	English	Accessible	Included	Yes	Included
11		Scopus	3	2	2	2	English	Accessible	Excluded	Yes	Included
12		Scopus	1	3	3	3	English	Accessible	Included		Included
13		Scopus	0	1	1	1	English	Accessible	Excluded		Excluded
14		Scopus	0	1	1	1	English	Accessible	Excluded		Excluded
15		Scopus	0	1	0	1	English	Accessible	Excluded		Excluded
16		Scopus	0	4	3	4	English	Accessible	Included		Included
17		Scopus	0	3	3	2	English	Accessible	Included	Yes	Included
18		Scopus	0	3	3	3	English	Accessible	Included		Included
19		Scopus	0	1	1	1	English	Accessible	Excluded		Excluded
20		Scopus	0	1	2	1	English	Accessible	Excluded		Excluded
21		Scopus	0	3	4	3	English	Not Accessible	Excluded		Excluded
22		Scopus	0	1	3	1	English	Accessible	Excluded	Yes	Excluded
23		Scopus	0	3	3	3	English	Accessible	Included	Yes	Included
24		Scopus	0	3	4	4	English	Accessible	Excluded	Yes	Included
25		Scopus	0	1	1	1	English	Accessible	Excluded		Excluded
26		Scopus	0	0	0	0	English	Accessible	Excluded		Excluded
27		Scopus	3	3	3	3	English	Accessible	Included	Yes	Included
28		Scopus	0	3	3	3	English	Accessible	Included		Included
29		Scopus	0	3	2	3	English	Accessible	Excluded		Excluded
30		Scopus	0	3	3	3	English	Accessible	Included	Yes	Excluded
31		Scopus	0	1	1	1	English	Accessible	Excluded		Excluded
32		Scopus	0	1	1	1	English	Accessible	Excluded		Excluded
33		Redalyc	0	3	3	3	English	Accessible	Included		Included
34		Redalyc	0	0	0	0	English	Accessible	Excluded		Excluded
35		Redalyc	0	0	0	0	English	Accessible	Excluded		Excluded
36		Redalyc	0	0	0	0	English	Accessible	Excluded		Excluded
37		Redalyc	0	3	1	3	Spanish	Accessible	Excluded	Yes	Excluded
38		Redalyc	0	0	0	0	Spanish	Accessible	Excluded		Excluded
39		Redalyc	0	0	0	0	Spanish	Accessible	Excluded		Excluded
40		Redalyc	4	0	0	0	Spanish	Accessible	Excluded		Excluded
41		Redalyc	0	0	0	0	Spanish	Accessible	Excluded		Excluded
42		Redalyc	4	2	2	2	Spanish	Accessible	Excluded		Excluded
43		Redalyc	0	0	0	0	Spanish	Accessible	Excluded		Excluded
44		Redalyc	0	0	0	0	Spanish	Accessible	Excluded		Excluded
45		Scielo	2	0	0	0	Spanish	Accessible	Excluded		Excluded
46		Scielo	0	0	0	0	English	Accessible	Excluded		Excluded
47		Scielo	0	0	0	0	Spanish	Accessible	Excluded		Excluded
48		Scielo	0	0	0	0	English	Accessible	Excluded		Excluded
49		Scielo	1	0	0	0	English	Accessible	Excluded		Excluded
50		Scielo	1	0	0	0	Spanish	Accessible	Excluded		Excluded
51		Scielo	0	0	0	0	English	Accessible	Excluded		Excluded
52		Scielo	0	0	0	0	Spanish	Accessible	Excluded		Excluded
53		Scielo	0	0	0	0	Spanish	Accessible	Excluded		Excluded
54		Scielo	0	0	0	0	Spanish	Accessible	Excluded		Excluded
55		Scielo	0	0	0	0	Spanish	Accessible	Excluded		Excluded
56		Scielo	0	0	0	0	Spanish	Accessible	Excluded		Excluded

T _r	Article	Source	Relevance	Reporting	Rigor	Credibility	Language	Accessibility	Status	Second evaluation	Final Status
57		Scopus	4	1	2	2	English	Accessible	Excluded		Excluded
58		Scopus	1	1	1	1	English	Accessible	Excluded		Excluded
59		Scopus	4	1	3	1	English	Accessible	Excluded		Excluded
60		Scopus	2	2	2	2	English	Accessible	Excluded		Excluded
61		Scopus	4	3	3	3	English	Accessible	Included	Yes	Included
62		Scopus	4	2	1	2	English	Accessible	Excluded		Excluded
63		Scopus	4	1	1	1	English	Accessible	Excluded		Excluded
64		Scopus	4	1	2	1	English	Accessible	Excluded		Excluded
65		Scopus	4	1	1	1	English	Accessible	Excluded		Excluded
66		Scopus	1	0	0	0	English	Accessible	Excluded		Excluded
67		Scopus	2	3	3	3	English	Accessible	Excluded	Yes	Included
68		Scopus	3	2	2	2	English	Accessible	Excluded	Yes	Excluded
69		Scopus	1	1	0	1	English	Accessible	Excluded		Excluded
70		Scopus	0	1	2	1	English	Accessible	Excluded		Excluded
71		Scopus	3	2	3	2	English	Accessible	Excluded	Yes	Excluded
72		Scopus	4	0	0	0	English	Accessible	Excluded		Excluded
73		Scopus	4	3	1	3	English	Accessible	Excluded	Yes	Included
74		Scopus	1	1	1	1	English	Accessible	Excluded		Excluded
75		Scopus	3	0	0	0	English	Accessible	Excluded		Excluded
76		Scopus	3	2	2	2	English	Accessible	Excluded	Yes	Excluded
77		Scopus	3	1	2	1	English	Accessible	Excluded		Excluded
78		Scopus	3	2	2	2	English	Accessible	Excluded	Yes	Excluded
79		Scopus	4	1	2	1	English	Accessible	Excluded	Yes	Excluded
80		Scopus	0	0	0	0	English	Accessible	Excluded		Excluded
81		Scopus	3	2	2	1	English	Accessible	Excluded	Yes	Excluded
82		Scopus	1	2	2	1	English	Accessible	Excluded		Excluded
83		Scopus	1	1	1	1	English	Accessible	Excluded		Excluded
84		Scopus	0	2	2	1	English	Accessible	Excluded		Excluded
85		Scopus	0	3	3	2	English	Not Accessible	Excluded		Excluded
86		Scopus	0	0	0	0	English	Accessible	Excluded		Excluded
87		Scopus	4	0	1	0	English	Accessible	Excluded		Excluded
88		Scopus	4	0	1	0	English	Accessible	Excluded		Excluded
89		Scopus	4	0	0	0	English	Accessible	Excluded		Excluded
90		Scopus	3	1	1	1	English	Accessible	Excluded		Excluded
91		Scopus	0	0	0	0	English	Accessible	Excluded		Excluded
92		Scopus	4	1	2	1	English	Accessible	Excluded		Excluded
93		Scopus	4	3	3	3	English	Accessible	Excluded	Yes	Included
94		Scopus	2	1	1	1	English	Accessible	Excluded	Yes	Excluded
95		Scopus	3	3	2	2	English	Accessible	Excluded	Yes	Excluded
96		Scopus	2	2	1	1	English	Accessible	Excluded		Excluded
97		Scopus	1	2	2	2	English	Accessible	Excluded	Yes	Excluded
98		Scopus	0	0	0	0	English	Accessible	Excluded		Excluded
99		Scopus	3	1	2	1	English	Accessible	Excluded	Yes	Excluded
100		Scopus	1	3	3	3	English	Not Accessible	Excluded		Excluded

9. Eligibility

During the data extraction phase, the included articles were read in full and analyzed according to the following descriptors, facilitating the integration, synthesis, and comparison of the data within the study.

- Research field
- Research type
- Stakeholders
- Form of Representation Proposed
- Communication intention of visual graphics
- Limitations
- Visual Principles
- Tool / Medium
- Evaluation principles of initial examples
- Findings on visualization
- Research Questions Analysis

Additionally, Google AI Studio served as an auxiliary tool, constrained by system instructions based on our theoretical framework, SLR protocol, and DE criteria. Rather than working independently, the AI processed and organized the researcher's preliminary synthesis, which the authors further interpreted to summarize results and formulate speculative propositions.

9.1 System Instructions (AI Studio - Persona)

The following system prompt was employed to define the AI persona within AI Studio:

You function as an auxiliary computational tool for an expert academic researcher conducting a Systematic Literature Review (SLR). Your role is strictly constrained to the refinement of the manuscript and the articulation of research analysis. The integration of this technology is strictly constrained to the refinement of the manuscript; the AI model is employed exclusively to rectify grammatical inconsistencies and to complement the analysis of the selected literature based on the specific, rigorous prompts and system instructions formulated by the research team.

You are NOT an independent generator of research insights. You function as a computational interface for processing, organizing, and articulating the analytical synthesis derived from the researcher's interpretation of the literature. Your outputs must remain anchored in the core theoretical principles and analytical interpretations provided by the user. Do not claim independent agency for insights; your output is a reflection and articulation of the user's analytical focus, elevated to a high-level academic standard.

You act as a specialized academic writing and structural assistant, enabling the researcher to express ideas with the voice and rigor of a senior scientific reviewer, doctoral advisor, and interdisciplinary theorist. Your analytical style must combine scientific precision, conceptual depth, methodological rigor, theoretical sensitivity, and advanced visual-literacy analysis.

The primary research context of this review is:

The investigation of metaphorical and multisensory visualization strategies for embodied carbon (EC) communication in architecture, especially for non-expert stakeholders.

Theoretical references that should implicitly guide your reasoning include:

- Juhani Pallasmaa (multisensory perception and architectural experience)
- Information Visualization theory
- Cognitive Fluency
- Cognitive Load Theory
- Cross-modal perception
- Data aesthetics
- Human-centered visualization
- Evidence-based design
- Visual metaphor theory
- Representation theory in architecture and art

Your task is to structure and refine the analysis I give of each article according to the following categories:

1. ARTICLE CHARACTERIZATION

Identify: Research type, Research field(s), Main research focus, and Stakeholders / target audience.

2. REPRESENTATIONAL ANALYSIS

Identify: Type of representation proposed, communication intention (what is being communicated, through which representational system, for which audience), type of data represented, and representational framework/application.

3. VISUAL PRINCIPLES

Identify and critically discuss: visual elements employed and perceptual/cognitive principles (cognitive fluency, cognitive load reduction, pattern recognition, etc.). Discuss metaphorical or cross-modal strategies.

4. TOOL / MEDIUM

Identify visualization technologies, digital systems, AI tools, computational frameworks, interactive platforms, or representational media used.

5. EVALUATION ANALYSIS

Critically examine evaluation type, evaluation principles, and methodological robustness. Distinguish clearly between empirically validated findings and conceptual/theoretical claims.

6. OUTCOMES / FINDINGS

Analyze effective strategies, graphical limitations, representational modifications, and new visual directions. Focus on shifts from abstract to perceptual representation, sensory engagement, contextual interpretation, and accessibility for non-expert audiences.

7. RESEARCH QUESTIONS ANALYSIS

Answer the following questions in sophisticated academic prose paragraphs:

QUESTION 1: What class and type of limitations are documented regarding data visualization, and what principles are explored to address those limitations? (Focus on abstraction, cognitive overload, stakeholder segmentation, lack of contextual meaning, representational fragmentation, expert-oriented communication, and perceptual limitations).

QUESTION 2: What existing precedents of metaphorical or non-abstract data representation exist in architecture, graphic design, or other disciplines that can inform and substantiate possible visual directions for LCA representation? (Focus on metaphorical systems, artistic precedents, perceptual representation, data aesthetics, multisensory or cross-modal approaches, contextual visual systems, and emotional encoding).

QUESTION 3: Which graphic limitations and principles were discussed in terms of altering the intended target audience that has access to the communicated data? (Focus on democratization of information, accessibility, interpretability, communication for non-experts, intuitive representation, stakeholder inclusion, and perceptual intelligibility).

WRITING STYLE REQUIREMENTS

- Resemble a high-level peer-reviewed academic analysis.
 - Maintain conceptual sophistication and methodological neutrality.
 - Avoid simplistic summaries and repetitive wording.
 - Prioritize interpretation over description, and conceptual relationships over isolated observations.
 - Do not fabricate information. If an article only implies something indirectly, explicitly state that it is implicit or inferred.
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9.2 Data Extraction

General Approach

ID / Article Title	03 / DATA VISUALIZATION AS ARTISTIC EXPRESSION
Year	2025
Research type	Theoretical
Research field	Data art
Stakeholders	Inexisted
Representation Proposed	It's not proposed. Data Art is discussed.
Communication Intention of Visual Graphics	To transform data visualization from purely informational representation into an expressive visual experience capable of generating emotional, perceptual, and interpretative engagement.
Visual Principles	Color (for mood/hierarchy), texture (simulated or density-based), composition (symmetry/asymmetry/rhythm), and geometry (structural accuracy) vs. organic forms (biological fluidity).
Tool/Medium	Creative coding frameworks (Processing, p5.js, D3.js, TouchDesigner), AI/GANs for generative aesthetics, linking analog sketches/physical artifacts with digital processing like CNC/3D printing).
Evaluation	Not developed
Findings on type of visualization	The study validates the transition from "passive representation" to "dynamic, participatory, and multisensory productions." Visualization can function as artistic expression and aesthetic and emotional dimensions improve interpretability and engagement.
RQ1	It is documented a "passive representational" limitation, where data is displayed merely to be read, failing to communicate broader cultural, environmental, or personal significance. This is identified as an "affective gap"—the failure of purely analytical graphics to provoke engagement. To address that the principle of "Interpretative Clarity" is introduced as a necessary counterbalance to the risk of "excessive abstraction," ensuring that while the representation is metaphorical or artistic, it remains grounded in meaningful and organized relationships.
RQ2	It cites historical scientific illustration (da Vinci, Vesalius) and cartography, which balanced empirical accuracy with "ornamentation" and "symbolic decoration." Also, the influence of the Bauhaus movement in using line, color, and geometry to express the "generative aesthetic" of structural data. This suggests that EC representation can rely on using

	texture and form to evoke the "weight" and "impact" of specific materials. The article also points to "environmental and weather-based sculpture" (Kim et al., 2020) as a precedent for physicalizing data, offering a pathway for EC visualization to become an immersive, spatial experience rather than a digital report.
RQ3	The current graphic limitations—specifically high abstraction and "expert-oriented" complexity—exclude general audiences, thereby limiting the social impact of data. It identifies that "high abstraction may confuse general audiences", creating a barrier to participation. To democratize access, the authors propose a move toward "human-centered data storytelling" by utilizing familiar aesthetic languages—such as natural patterns, organic mappings, and intuitive symbols—the visual systems attempt to "convert ordinary communal spaces into spaces of collective thought."

ID / Article Title	05 / Fluidly Revealing Information: A Survey of Un/foldable Data Visualizations
Year	2025
Research type	Systematic Literature Review and Classification
Research field	Information Visualization
Stakeholders	HCI researchers, visualization designers
Representation Proposed	Fluid, transformative interfaces that utilize geometric and semantic "un/folding" to manage abstraction layers.
Communication Intention of Visual Graphics	To facilitate the progressive revelation of complexity data, enabling users to explore hierarchical or complex systems without losing their spatial or contextual orientation.
Visual Principles	Animation, geometric deformation (fanning, stretching, sliding), hierarchical layering, semantic abstraction.
Tool/Medium	It doesn't refer to one specific. Cross-platform, as web-based catalogs, interactive prototypes and Augmented Reality/XR.
Evaluation	Not developed. Taxonomic synthesis.
Findings on type of visualization	It is identified ten "families" of un/folding (Fanning, Hinging, Nesting, Scrunching). These provide a "menu" of interaction strategies for designers.
RQ1	The article identifies the limitation as the conflict between "data complexity" and "available screen real-estate," which often leads to cognitive overload when users are confronted with massive, static datasets. To overcome these limitations, the article explores the principle of Fluid Interaction and Visual Momentum. By utilizing un/folding, the interface

	allows for "gradual transitions" that permit the user to maintain a sense of context. The principle of Context Preservation is central here; it ensures that the "base visualization" remains an anchor point, preventing the user from becoming disoriented.
RQ2	The physical metaphors of unfolding can transform the abstract carbon metric into a spatial interaction, making the data "felt" through the act of manipulating the object, rather than just reading a static percentage. These are not merely analogies; they are structural templates. For EC communication, we could adapt the "Nesting" family (like Matryoshka dolls) to represent a building as a nested hierarchy where, as a component like a wall is unfolded, its material layers and associated carbon values are revealed. The "Fanning" mechanism, which distributes elements radially, could be used to represent the lifecycle of a material, where the "fan" spreads out to reveal different stages of carbon impact.
RQ3	It identifies a limitation in existing visualization tools: they are often designed for analytical "truth" rather than human engagement, which creates a barrier for non-expert stakeholders. The report documents that "pleasurable experiences"—characterized by immediate, fluid feedback—are crucial for non-expert engagement. By incorporating principles from industrial design (collapsibles) and nature, these visualizations become more intuitive and "delightful." This alters the audience dynamic: instead of requiring the stakeholder to learn how to decode a professional graph, the "un/folding" interface invites them to play with the data, enabling the democratization of complex information.

ID / Article Title	08 / Improve Usability of Multidimensional Data Exploration with Water Fountain Based 3D User Interface Metaphor
Year	2024
Research type	Empirical / Proof-of-concept (the validation of a prototype)
Research field	Human-Computer Interaction (HCI), Information Visualization
Stakeholders	Data analysts, public health decision-makers
Representation Proposed	Metaphorical 3D spatial interface. The model maps multivariate data of epidemiology datasets onto the physical behavior of a fountain (height, flow, geysers, environmental climate cues).
Communication Intention of Visual Graphics	To facilitate cognitive associations, thereby reducing the mental effort required for multi-dimensional data exploration compared to traditional 2D interfaces like excel.
Visual Principles	Height (severity), color (category/density), water-sprout density, and temporal indicators (sun/cloud for climate variables).
Tool/Medium	Unity 3D game engine (for rendering 3D spatial interface), Leap Motion (gesture-based interaction), C# (for flow/behavior modeling).
Evaluation	Controlled experimental study with four groups (n=40) evaluating the usability measured via the Computer System Usability Questionnaire (CSQU), task completion metrics, and time-to-completion.

Findings on type of visualization	The research validates that physicalizing data into 3D metaphors (fountain sprouts) significantly improves "intuitive" data navigation for non-specialists compared to grid-based 2D tools.
RQ1	The article identifies the primary limitation as "cognitive overload" resulting from traditional 2D multidimensional visualization paradigms like pivot tables. These representations are disconnected from intuitive human perception, forcing analysts to rely on high-effort cognitive processing. When data dimensionality increases, the traditional "dashboard" model fails to scale effectively, leading to visual clutter and navigation difficulty. To address this, the authors explore the principle of Cognitive Metaphor Theory, which posits that the human brain operates through pre-existing physical schemas. By designing a system that mirrors the natural behavior of a water fountain, the researchers attempt to bypass the slow, effortful cognitive processing required for abstract data. The core principle explored here is the delegation of information processing to the perceptual system; by framing data points as physical, height-variable sprouts in a 3D fountain.
RQ2	The article cites several precedents for metaphor-based 3D visualization, such as the "Solar System" metaphor for social networks and the "VirtualDesk" or "Space-Time Cube" for spatiotemporal data. These precedents support a transition toward "immersive ecosystem" metaphors. Much like how the water fountain uses height to represent patient severity, an EC visualization could use the "Fountain" or "Growth" metaphor to represent carbon flux: where the height or volume of a material node in a 3D building model corresponds directly to its embodied energy. The precedent idea of "environmental sculpture" or "spatial mapping" can be applied to EC, and its data could be represented as a 3D landscape where architectural volumes deform or change visual properties (color/texture) based on their environmental impact.
RQ3	The article addresses the democratization of complex data by attacking the barrier of "high cognitive effort." It identifies that standard tools—like MS Excel—are restrictive, accessible only to those with high data-literacy and training. The graphic principle explored to democratize this information is "Intuitive Learnability." By using a water fountain—a universally recognizable physical object—the interface removes the requirement for specialized training, allowing stakeholders to grasp the "severity" or "density" of data immediately upon viewing. This shifts the audience from "expert data analysts" to a more inclusive set of "decision-makers."

ID / Article Title	09 / Designing complex network visualisations using the wayfinding map metaphor
Year	2024
Research type	Design-oriented research
Research field	Information Visualization, HCI
Stakeholders	domain experts (like molecular biologists) and designers
Representation Proposed	A network diagram augmented by "wayfinding" navigational structures (landmarks, paths, districts).

Communication Intention of Visual Graphics	To transform the complex network into an explorable "landscape," thereby using the user's existing mental model of geographic map navigation.
Visual Principles	Nodes, Edges, Clusters (districts), Landmarks (personalized markers), and visual "paths" (connections).
Tool/Medium	ForceAtlas2 layout algorithm; Python MiniSom (for dimension reduction).
Evaluation	Design-based research with stakeholder involvement (workshops and qualitative feedback).
Findings on type of visualization	Successful implementation of "Landmarks" (search/flagging) to help users regain orientation in dense networks.
RQ1	The article identifies the fundamental limitation of network visualization as the "hairball effect," where a high density of visual elements and non-linear relationships leads to total loss of user orientation. To address these limitations, the researchers leverage the principle of "Wayfinding Semiotics." They argue that if network diagrams are treated as terrains, the design problem shifts from "how to display all data" to "how to provide orientation, exploration, and navigation." By mapping wayfinding tasks—specifically orientation, exploration, and navigation—to interactive network operations, they provide a structured principle for interaction design.
RQ2	The article relies heavily on the precedent of cartography and urban design (Lynch's landmark theory) as a source for mental models. This substantiates the idea that complex datasets, including Embodied Carbon (EC), can be effectively represented using spatial and navigational metaphors. The precedent proves that mapping non-spatial data (genetics) onto spatial metaphors (maps) is not only possible but cognitively superior for exploration.
RQ3	The article argues that the graphic limitation for non-experts is the reliance on technical, abstract, and static representations that assume prior "visual literacy" in graph theory. By utilizing the wayfinding metaphor, the researchers advocate for a democratization of data. They posit that because "ordinary users" have a deeply ingrained mental model of maps, this design choice drastically lowers the learning curve. In an architectural or environmental context, this principle of "low-threshold interaction" is vital. By designing an EC communication system that mimics a wayfinding map, architects can empower non-expert stakeholders (clients, the public) to participate in decision-making without requiring deep environmental science knowledge. The principle here is to trade "analytical purity" for "navigational intuitiveness," where the visualization system acts as a host to the user's natural exploratory impulses rather than a rigid data-audit tool.

ID / Article Title	10 / Visually encoding orthogonal planar graph drawings as graph mazes: An eye tracking study
Year	2024
Research type	Experimental (Comparative eye-tracking study)

Research field	Information Visualization, HCI
Stakeholders	Network visualization researchers
Representation Proposed	The "Graph Maze" metaphor, which transforms abstract network connections into a street-like, enclosure-based spatial structure. Nodes become "street segments" and edges become "streets," with empty space serving as "walls/enclosures."
Communication Intention of Visual Graphics	To facilitate more intuitive path-finding by leveraging pre-existing human familiarity with navigation and spatial enclosure.
Visual Principles	Enclosure lines, street segments, dead-ends, junctions (nodes), and spatial blocks (empty space). Geodesic path tendency and Visual Orientation/Reorientation.
Tool/Medium	Tobii Eye Tracker 5 (eye-tracking hardware), Tobii Pro Lab (analysis software), and programmatic generation of planar orthogonal graphs (via de Fraysseix's algorithm) and Google Drawings for stimuli presentation.
Evaluation	Task completion time, error rates in a comparative user study (N=26).
Findings on type of visualization	Enclosure-based metaphors (mazes) are a viable alternative to connection-based node-link diagrams.
RQ1	The article addresses the limitation of traditional "connection-based" network diagrams (node-link diagrams). Specifically, in dense or complex graphs, the eye struggles to trace lines (edges) through a chaotic field, a limitation often exacerbated by edge-crossing or non-linear layouts. The principle explored to address this is the shift from "connection" to "enclosure." By re-encoding the graph as a "maze," the design leverages the principle of naturalistic spatial navigation. This effectively turns the data structure into an architectural space. The cognitive principle here is the utilization of human spatial intelligence to solve the abstract problem of path-finding in data.
RQ2	The precedent identified in this study is the "street-like visual metaphor." For Embodied Carbon (EC) representation, this is a highly potent precedent. This architectural precedent allows for a system where a user "walks" through the carbon footprint of their design. It is a transition from an "abstract dashboard" to a "navigable environment," where the user's task is to navigate the maze of material impacts to find the "exit" of the lowest-carbon, high-performance configuration.
RQ3	The article discusses limitations regarding "prior knowledge" and "experience levels," noting that traditional node-link diagrams are abstractions that often alienate non-expert users. The study demonstrates that when data is encoded in a way that aligns with familiar life experiences (like a maze or city street), the dependence on specialized graph-reading skills decreases. This has profound implications for democratizing data: if one designs an interface where the visual metaphor is "street-based" rather than "node-based," the intended audience expands from specialists to the general public. By adopting enclosure-based representations that feel "natural" and "street-like," architects and designers can make complex environmental data legible to non-experts, allowing them to engage with the system intuitively without needing to understand the underlying graph-theoretical structures.

ID / Article Title	11 / Pacemod: parametric contour-based modifications for glyph generation
Year	2023
Research type	Technical Methodological
Research field	Computer Graphics, Visualization Design
Stakeholders	Visualization designers, interface architects
Representation Proposed	Parametric, contour-based metaphoric glyphs, a generative approach that manipulates the geometry (using diffusion curves) and color attributes of a base icon to create multivariate data.
Communication Intention of Visual Graphics	To resolve the conflict between automated data representation (unrecognizable) and metaphoric representation (intuitive). The intention is to enable data-driven modification of icons that the viewer instinctively understands.
Visual Principles	Contour modification (geometric translation), color modulation, wave amplitude, and frequency.
Tool/Medium	Diffusion Curve Image (DCI) representation, cubic B-splines, Distance Transforms (for intersection prevention).
Evaluation	Technical validation of the parametric method and exploratory user study.
Findings on type of visualization	The study confirms that periodic contour modification is an effective strategy for uncertainty encoding.
RQ1	The limitation explored is related to the difference between manual, metaphoric design (high effort, high intuitivity) and automated, data-driven glyph generation (low effort, low recognizability/intuitivity). The current limitations of automated approaches are that they often yield abstract shapes that lose the semiotic connection to the real-world object they represent. To bridge this, the authors introduce the principle of "parametric contour-based modification." This principle allows for the retention of an icon's "base identity" while enabling the geometry itself to serve as a variable data container. This preserves the iconic "anchoring" of the visualization (the what) while encoding the data via shape distortion (the how much), effectively managing cognitive load by ensuring that the user does not have to learn a new, abstract visual syntax to understand the represented data.
RQ2	The article cites precedents such as "corn-cob glyphs" (for agricultural data) and "leaf-like glyphs" (for forest fire monitoring). These examples demonstrate the power of mapping ecological variables directly onto the morphology of the object associated with the domain. For Embodied Carbon (EC) representation the PACEMOD framework could be used to programmatically distort the silhouette of a steel I-beam or a brick icon to represent its carbon intensity. The contour frequency could encode the carbon value, while the "base shape" remains instantly recognizable to an architect.

RQ3	The article focuses on increasing the "intuitivity" and "learnability" of data representations, which is a key driver for accessibility among non-experts. The graphic limitation identified is that non-metaphoric glyphs often require external legends or significant training to be "read" correctly. By using familiar base icons—"well-understood visuo-spatial phenomena"—the authors argue that the barrier to entry is lowered significantly and designers can create accessible, metaphoric visual languages at scale, democratizing the consumption of complex datasets by ensuring that the "picture becomes the thing it represents," thereby bypassing the need for advanced visual literacy in the target audience.
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ID / Article Title	12 / Visualization Metaphors in the Tasks of Exploratory Analysis of Heterogeneous Data
Year	2024
Research type	Methodological / Prototyping
Research field	Information Visualization, Cyber-Physical Systems
Stakeholders	Data analysts
Representation Proposed	3D metaphoric visualization. The system uses a "spatial metaphor" (cylindrical coordinates/parallel planes) to organize data and "representation metaphors" (markers, petal glyphs, object profiles) to encode properties.
Communication Intention of Visual Graphics	To enhance the analyst's cognitive potential by mapping multi-dimensional attributes into a coherent visual space where relationships and anomalies are intuitively "visible."
Visual Principles	The "spatial relationship metaphor" is the primary driver, using a cylindrical system to create a "common visual model." Radial distance (position on concentric circles), geometric shapes (tetrahedrons, cubes), color-coding (object identity/deviation magnitude), and profile profiles (broken lines/petals).
Tool/Medium	NET Framework/WPF (Windows Presentation Foundation) for the software tool.
Evaluation	Proof-of-concept (POC). It does not contain a formal usability study with human subjects, but rather a functional validation of the framework's ability to reveal anomalies and manage data gaps.
Findings on type of visualization	The "shared visual field" strategy—integrating different metaphorical views simultaneously—was found to improve cognitive clarity.
RQ1	The article addresses the limitations of visualizing "heterogeneous data," characterized by weak formalization, high abstraction, and data incompleteness (missing values). The primary limitation is the cognitive bottleneck of the analyst: traditional 2D or static representations fail to capture the multi-dimensional, irregular nature of these datasets, leading to incomplete or erroneous interpretations. To address this, the authors explore the principle of "Cognitive Clarity." They posit that by transferring characteristics of the

	data into a 3D "visual model space," they can delegate the processing task to the human perceptual system, which is faster than the cerebral cortex. The technology employs iterative cycles of "perception" and "interactive model control," allowing the analyst to manipulate the visual image to gain insight.
RQ2	The article draws on the precedent of "spatial relationship metaphors," which are common in architectural and urban planning visualizations (e.g., mapping movement trajectories). The "petal metaphor" (a variant of radar/spider charts) is another precedent that translates well to Life Cycle Assessment (LCA). For EC communication, there is a visual direction where material properties are not presented in a table, but as "petals" of a glyph, where the area of each petal corresponds to a specific environmental impact category. The study also substantiates the "deviation from reference" metaphor. In LCA, this could be used to represent the environmental "atypicality" of a material choice against a "best-in-class" reference object.
RQ3	The article discusses the limitation of "weakly formalized tasks" which are often only accessible to highly skilled analysts. By utilizing visualization metaphors, the authors aim to increase the democratization of data exploration. The principle explored here is that "visual models should be simple and convenient to interact with." By embedding data into intuitive geometric structures—like tetrahedrons or petals—that possess immediate visual logic, the system attempts to lower the expertise barrier. This graphic strategy alters the audience dynamic by substituting technical data-literacy with perceptual visual-literacy.

ID / Article Title	16 / Visualizing ordered bivariate data on node-link diagrams
Year	2024
Research type	Experimental
Research field	Information Visualization, HCI
Stakeholders	Decision-makers
Representation Proposed	Bivariate edge-glyph design. The model uses a two-channel encoding: color (hue) for the primary attribute and central stripe patterns (frequency) for the secondary attribute.
Communication Intention of Visual Graphics	To resolve the "visual interference" inherent in common bivariate encodings (e.g., width-lightness, transparency-saturation), where encoding channels often conflict or mask one another. The intention is to achieve high precision and speed in dual-attribute identification.
Visual Principles	Hue (edge color) and Texture/Stripe Frequency (edge glyph).
Tool/Medium	NetworkX for layout generation; static image stimuli generation; PsychoPy/Prolific for psychophysical evaluation.

Evaluation	Two-stage controlled quantitative user study.
Findings on type of visualization	Stripe frequency (texture) is a superior secondary channel to lightness/transparency, which often cause visual masking or occlusion in node-link diagrams.
RQ1	The article documents a failure in bivariate network visualization: the "interference effect" between visual channels. Standard approaches (like width paired with lightness, or saturation paired with transparency) suffer from the problem that the visual properties mask each other, making them difficult for the eye to perceive simultaneously and accurately. The researchers explore the principle of "Visual Entropy" as a solution. By decoupling the channels—assigning the primary attribute to the edge's color (hue) and the secondary attribute to the edge's internal texture (stripe frequency)—they minimize visual interference. This principle of channel independence allows for higher accuracy in "dual target" visual search tasks, where the user must discern two data points on a single edge without one property degrading the perception of the other.
RQ2	The article references Bertin's semiology of graphics, specifically the "texture" channel, which provides a direct precedent for using repetitive patterns to represent ordered data. For Embodied Carbon (EC) representation, this is highly relevant. One could substantiate a visual direction where the primary edge color represents the carbon intensity of a material flow, while the "stripe frequency" (texture) represents the uncertainty or data quality of that LCA assessment. The precedent suggests that rather than relying on numeric labels—which are hard to read at a glance—we should use texture-frequency as a robust, non-interfering secondary channel to make EC data "feel" more reliable.
RQ3	The study focuses on the limitation of "cognitive burden" for decision-makers who must rapidly extract insights from dense graphs. By proving that the Vizent design leads to higher accuracy and efficient performance even in complex graphs, the authors address the accessibility gap for non-expert users who might struggle with traditional, subtle encodings (like transparency or lightness, which require fine-grained discrimination).

ID / Article Title	17 / Interactive Input and Visualization for Planning with Temporal Uncertainty
Year	2023
Research type	Empirical / Design-based Research
Research field	(HCI), Information Visualization
Stakeholders	Project planners, non-expert users engaged in personal scheduling
Representation Proposed	The "Tube" metaphor—a graphical evolution of Gantt and PlanningLines representations. It uses brackets, shaded bars, and gradients to represent temporal intervals and probability distributions.
Communication Intention of Visual Graphics	To facilitate the "externalization of implicit knowledge" regarding uncertainty. The intention is to enable users to intuitively sketch plans where uncertainty is not a hidden variable but a primary visual feature.

Visual Principles	Brackets (bounding intervals), dark bars (minimum duration), lighter extensions (buffer time), and gradients/textures (representing probability).
Tool/Medium	NET/WPF-based application
Evaluation	Quantitative user study (two tasks: drawing assignment and reading assignment).
Findings on type of visualization	The use of "Tube" configurations successfully allowed participants to externalize complex uncertainties (like "expected end" with probabilistic delays) without significant difficulty.
RQ1	The research explicitly documents the failure of traditional scheduling visualizations—such as Gantt charts and PERT diagrams—to account for the inherent "elasticity" of real-world events. The authors note that when uncertainty is included, it is often relegated to textual notes or static error bars, which users frequently misinterpret as high certainty. To resolve this, the researchers introduce the principle of "Direct Externalization." By enabling users to sketch their uncertainty—drawing the bounds of an activity in real-time—the visualization becomes a participatory act.
RQ2	The article positions its "Tube" metaphor as an evolution of PlanningLines and PERT charts, but the broader precedent here is the translation of "probabilistic duration" into "visual volume." This is highly relevant to Embodied Carbon (EC) representation, which is similarly riddled with uncertainty (e.g., EPD data ranges, material lifespan variability). Just as the "Tube" uses a dark core (minimum duration) and a shaded extension (buffer time) to show temporal uncertainty, an EC visualization could use a "Carbon Tube" to represent a material's impact. The core would show the most likely carbon footprint, and the shaded "buffer" would represent the range of uncertainty based on different manufacturing scenarios or regional data variations.
RQ3	The article addresses the democratization of scheduling by critiquing the high-threshold complexity of traditional project management tools. The limitation of these tools is that they require expert knowledge to configure, effectively locking out non-experts from the planning process. The principle of "low-threshold interaction" (sketching) alters the audience dynamic by removing the need for formal data entry. By allowing users to express uncertainty through the simple act of drawing, the tool lowers the bar for participation, making the planning process accessible to individuals who are not project management professionals.

ID / Article Title	18 / The Role of Subjective Perception in Visualization, Controllability Hypothesis and Abstract Visualization
Year	2022
Research type	Theoretical
Research field	Information Visualization, Cognitive Science

Stakeholders	Visualization theoreticians
Representation Proposed	Abstract vs. Object-oriented visual models. The authors advocate for models that function as a "linguistic space" rather than mere pictorial representations.
Communication Intention of Visual Graphics	To provoke the user, initializing a "cognitive search" where the visualization does not simply "inform," but actively supports the user in generating new knowledge or hypotheses.
Visual Principles	Direction (top-bottom/left-right), Time-scale control, Rhythm (repetition/structure), and Perspective/Wide-angle distortion.
Tool/Medium	Autodesk 3ds Max (with MaxScript and Python)
Evaluation	Proof of concept via social data analysis.
Findings on type of visualization	Manipulation of spatial "ergonomics" (rhythm, direction, perspective) can be used as a controlled design parameter to influence user perception.
RQ1	The article documents a fundamental limitation in the "unformed apparatus" of visual communication, characterized by the subjectivity of interpretation and the lack of a formal basis for why certain visualizations work. The primary limitation is the "passive perception" or "visual informing" state, where the user treats the visualization as a static, indisputable symbol rather than an active tool for exploration. This leads to what the authors call a "negative cognitive effect," where users substitute dynamic analysis for a quick comparison with a flawed mental template. To address these limitations, the authors explore the principle of "Perceptible Controllability." This principle suggests that visualization systems should not just show data; they should intentionally influence the user's role in the "user-space" system. By applying principles of visual ergonomics—specifically rhythm, directionality, and perspective control—designers can force users out of passive consumption and into active cognitive search.
RQ2	The article provides a robust theoretical precedent for "abstract environments" as generators of new hypotheses. In the context of EC communication, this suggests we should not just aim for "realistic" material representations, but also for "abstract" spaces that force architects to rethink their carbon mental models. The paper's discussion of "rhythm" and "perspective control" suggests that if EC data were visualized in a 3D, transformable space, designers could use perspective distortion to emphasize certain phases of a material's life cycle.
RQ3	The paper discusses the limitation that personal user experience is "extremely extensive and diverse," making the creation of a "universal" metaphor difficult. This limitation implies that static, universal visualizations are bound to fail. To overcome this, the authors discuss the principle of "adaptive visualization," where the visualization parameters are adapted to the specific characteristics of the user or group. By framing visualization as an "ergonomic" task—designing the visual space to match the user's perception capabilities—the intended audience shifts from a monolithic "user" to a dynamic set of individuals whose behavior and goals the system must influence.

ID / Article Title	23 / The State of the Art in Empirical User Evaluation of Graph Visualizations
Year	2021
Research type	Systematic Literature Review
Research field	Information Visualization, Human-Computer Interaction (HCI)
Stakeholders	Inexisted
Representation Proposed	The article does not propose a new representation but critiques and classifies existing ones: Node-link diagrams, adjacency matrices, and hybrid approaches.
Communication Intention of Visual Graphics	To shift the focus of visualization research from "aesthetic representation" to "human performance," establishing what visual encodings actually work for specific user tasks (e.g., path-tracing vs. cluster detection).
Visual Principles	Visual elements employed: Nodes (shape, size), edges (width, curvature, partial links, bundling), and layout algorithms (force-directed, radial, hierarchical).
Tool/Medium	Various
Evaluation	Review analysis
Findings on type of visualization	Straight-line edges are generally superior to curved ones for path-tracing tasks. Tapered edges and animated transitions perform better than complex, multi-cue directed symbols.
RQ1	The article documents critical limitations stemming from "visual clutter" and the failure of existing layouts to facilitate rapid extraction of insight. A recurring limitation is the "hairball" structure, where high graph density renders node-link diagrams unreadable, leading to high cognitive load and inaccurate path tracing. To address these, the review identifies principles such as mental map preservation and clutter reduction. A fundamental finding is that users rely on a "geodesic path tendency"—they expect the shortest route to be visually direct (straight lines). Principles that violate this—such as excessive link curvature or non-intuitive layouts—impede the user's ability to "read" the data structure.
RQ2	The article discusses the "street-like visual metaphor" (as seen in metro maps or schematic cartography) as a precedent for representing non-planar or complex relational data. For Embodied Carbon (EC) visualization, this suggests a powerful visual direction: treating the lifecycle of a material not as a standalone data point, but as a "path" within an urban-like network. The precedent of using "partial links" and "closure" (gestalt principles) also suggests that we do not need to show every single connection in a material network to achieve intelligibility.
RQ3	The article addresses the democratization of graph data by critiquing studies that rely on "expert" users and "laboratory" conditions. It posits that the true test of a visualization's success is its performance with end-users. The graphic limitation identified is that highly

	"aesthetically pleasing" layouts (e.g., curved links) are often inferior in performance to "plain" layouts (straight links). This has major implications for the target audience: if the goal is to make EC data accessible to builders, architects, or public planners (non-experts), we must prioritize readability over aesthetic complexity.
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ID / Article Title	24 / Visualization of Emergency Needs Posted on Social Media by Metaphor Map
Year	2020
Research type	Design-based research
Research field	Human-Computer Interaction (HCI), Information Visualization, Social Network Analysis.
Stakeholders	decision-makers
Representation Proposed	A cartographic metaphor ("Metaphor Map") which projects abstract, non-spatial thematic data onto a familiar geographic-like terrain.
Communication Intention of Visual Graphics	To resolve the "fragmentation" of emergency data by unifying theme structure and evolutionary dynamics into a single, cohesive visual landscape, making complex demand patterns intelligible for rapid decision-making.
Visual Principles	Area (popularity/demand size), border thickness (categorical distinction), and line weight (flow of influence/social connection).
Tool/Medium	Web-based interactive mapping platform; Dynamic Topic Modeling (DTM); Single-Pass clustering algorithms for real-time thematic classification.
Evaluation	Preliminary usability study using the NASA Task Load Index (NASA-TLX).
Findings on type of visualization	Spatializing thematic data as a terrain successfully enables the detection of crisis stages (e.g., brewing, outbreak, extinction) that are often obscured in raw text streams.
RQ1	The article identifies the primary limitation as the "cognitive overload" resulting from fragmented, text-heavy data representation during time-critical crisis management. Traditional reporting methods decouple the structure of emergency needs from their evolutionary timeline, forcing analysts to manually bridge this information gap. To address this, the research explores the principle of "Metaphorical Cartography." By structuring data within a map-based metaphor, the system addresses cognitive load by leveraging the user's pre-existing spatial schemas. This allows for the integration of multiple data facets—popularity, structure, and evolution—into a unified visual field. The core principle is "cognitive congruence": ensuring that the visualization's internal logic mirrors the user's natural spatial orientation strategies, thereby turning abstract demand data into a "navigable terrain" where patterns can be identified instantly.

RQ2	The article builds upon the precedent of "cartographic visualization of non-spatial data" (e.g., Skupin, 2000; Fabrikant, 2003). In an architectural LCA context, this is a highly significant direction. Embodied Carbon (EC) data is often treated as tabular or spreadsheet-based, which limits its utility in design. The precedent of a "Metaphor Map" suggests that EC could be represented as an "Environmental Terrain." One could define "Continents" of material impact (e.g., carbon-heavy structures vs. sequestering materials) and "Highways" of material flow. This provides a tangible precedent for mapping EC metrics not just as numbers, but as spatial territories within a design proposal, allowing architects to "traverse" the carbon impact of their choices through an intuitive, map-based interface.
RQ3	The article documents the limitation of "expert-only" data analysis—where complex topic models remain locked away in specialized statistical reports. The authors argue that this prevents key stakeholders (emergency management personnel) from accessing vital information in time. The graphic principle utilized to democratize this information is the "Intuitive Metaphorical Interface." By replacing dense statistical outputs with familiar cartographic landmarks, the tool alters the audience profile from "data scientists" to "crisis managers." For EC communication, this principle is vital: if LCA data is represented through simple, metaphoric maps rather than abstract technical charts, the intended audience expands to include the entire project team, including clients and contractors, who can then meaningfully participate in sustainability decisions without requiring advanced data-literacy training.

ID / Article Title	27 / Stroke-based splatting: an efficient multi-resolution point cloud visualization technique
Year	2017
Research type	Technical Methodology (Computer Graphics)
Research field	Computer Graphics, Non-Photorealistic Rendering (NPR), Human-Computer Interaction (HCI)
Stakeholders	Graphics researchers, 3D visualization developers
Representation Proposed	Impressionistic surface-aligned splatting (SAS). It translates raw, unstructured point cloud data into a series of "brush strokes" oriented by tangential vectors.
Communication Intention of Visual Graphics	To enhance the visual readability of sparse 3D data. By utilizing the cultural metaphor of painting, the authors intend to improve "shape perception" in situations where standard meshes or elliptical splats fail (due to noise or low resolution).
Visual Principles	Splat shape (brush stroke), color clustering (grouping), and coherent tangent vectors (orientation).
Tool/Medium	OpenGL, GLSL (for shader-based rendering), Point Cloud Library (PCL) for preprocessing.
Evaluation	Empirical comparative study (N=31, online questionnaire).

Findings on type of visualization	Brush-stroke orientation, based on coherent tangent vectors, is a superior strategy for communicating surface curvature compared to generic elliptical splats.
RQ1	The article documents critical limitations in standard point cloud rendering, specifically "background/foreground confusion" and the loss of definition in close-up or low-resolution scenarios. When sparse point clouds are rendered using traditional methods, the result is often "pixelated artifacts" that lack structural coherence. This creates a significant perceptual limitation: the eye cannot "read" the curvature or shape of the object. To address this, the authors explore the principle of Artistic Metaphorical Encoding. By shifting the paradigm from purely geometric representation to a "painterly" depiction, the principle of coherence is introduced. Through the application of the Householder formula for tangent estimation, the system ensures that individual "brush strokes" (splats) are oriented to reflect the underlying surface curvature, creating a visually continuous "flow" that guides the user's eye, thereby transforming raw noise into a readable, coherent form.
RQ2	The study leverages the precedent of Impressionist painting—an established cultural aesthetic—to substantiate its rendering choices. This is highly relevant to Embodied Carbon (EC) visualization. EC data, like raw point clouds, is often "noisy" or fragmented. The precedent here suggests that we should not aim for an "accurate" numerical visualization, but rather an "impressionistic" one. In an EC context, this means that for a building's design-stage assessment, where the data is necessarily uncertain, we could use an "impressionistic" style to communicate the overall carbon "atmosphere" of a design. If a material is carbon-intensive, the "brush stroke" might be darker or thicker; if it is low-carbon, it might be more refined. This precedent substantiates a shift away from high-fidelity (but often inaccurate) reporting toward expressive, stylized representations that communicate the "gestalt" or overall environmental performance of a design.
RQ3	The article discusses the limitation that traditional point-cloud visualization tools are "not suitable" for broad interaction because they require specialized knowledge to interpret artifacts or "fill in the gaps" mentally. By applying a cultural metaphor—Impressionist painting—the study broadens the potential audience. Because users are "culturally trained to interpret" this style, they can understand the geometry without needing a degree in computer graphics or point-cloud rendering. This is a crucial principle for EC communication: by using familiar artistic visual languages, we can remove the specialized technical barrier that usually surrounds LCA data. The principle of Cultural Familiarity allows the visualization to become an interface for everyone, enabling architects, clients, and even the public to engage with complex material data by relying on their innate ability to read visual "quality" rather than technical data.

ID / Article Title	28 / Multidimensional data exploration by explicitly controlled animation
Year	2017
Research type	Methodological
Research field	Information Visualization (InfoVis), Human-Computer Interaction (HCI).
Stakeholders	Data analysts, software engineers

Representation Proposed	A combination of "small multiples" (a grid of thumbnail views) and interactive animation paths that allow the user to traverse and interpolate between views.
Communication Intention of Visual Graphics	To create a cohesive "visual story" out of partial, disjointed data views. The intention is to enable users to "sketch" their own exploration path, allowing for continuous, meaningful transitions between different analytical perspectives of the same dataset.
Visual Principles	Thumbnails (small multiples), animation (smooth transitions), trajectories (the "path" of exploration), and locks (fixed elements in the visualization).
Tool/Medium	C#, .NET 4.5, OpenGL, CUDA (for real-time interpolation of up to one million points).
Evaluation	Exploratory application-based validation using real-world datasets (air traffic control, software maintenance, machine learning).
Findings on type of visualization	Explicit, user-controlled animation paths are more effective than automatic linear interpolation, as they allow users to define their own "visual storyline."
RQ1	The article identifies a key limitation in the fragmentation of high-dimensional data visualization, where the need for multiple views leads to an overwhelming "exploration space" that users struggle to synthesize. The cognitive limitation documented is the user's inability to maintain a coherent mental model when switching between disconnected partial views. To overcome this, the authors explore the principle of Visual Continuity. By treating the "view-space" as a navigable landscape rather than a collection of separate charts, they apply the principle of smooth interpolation to bridge cognitive gaps. The principle of "Explicitly Controlled Animation" allows the user to see the evolution of their data views, essentially treating the analytical process as a coherent "visual story" rather than a series of disparate snapshots, thereby significantly reducing the cognitive effort required to synthesize partial insights.
RQ2	The article discusses "projection boards" and "storytelling via small multiples" as key precedents. In the context of Embodied Carbon (EC) visualization, this suggests a visual direction based on "scenarios." If an architect is comparing material alternatives for a building, these alternatives could be organized on a "projection board" where each thumbnail represents a different material palette or assembly method. Instead of viewing these as separate, disconnected files, the user could "sketch a path" between them, using smooth animation to visualize how the embodied carbon footprint morphs as one material is substituted for another. This precedent substantiates a design where the architect can "animate" the carbon impact of their design choices, making the abstract consequences of material selection concrete and temporally traceable.
RQ3	The article discusses the limitation that high-dimensional data exploration is often a "black box" that requires expert knowledge to navigate. The authors highlight that the current reliance on parameter tuning (e.g., adjusting specific numbers to change a view) excludes non-experts. The principle explored to alter this is "Direct Manipulation and Visual Sketching." By allowing the user to simply drag a focus point or sketch a path on a grid of thumbnails, the system removes the requirement to understand the underlying mathematical parameters. This alters the target audience by making the exploration tool accessible to those who prioritize "visual interaction" over "parameter-based tuning." In the context of EC communication, this principle is vital: it suggests that we can make the complex "parameter space" of material carbon performance accessible to the architect or client by allowing them to "play" with the visualization through simple, direct, and visual interaction, rather than requiring them to understand the backend LCA calculations.

ID / Article Title	33 / Materializing data: notes on collaboration and tangible interfaces with excerpts and additions
Year	2017
Research type	Technical Methodology (Computer Graphics)
Research field	Computer Graphics, Non-Photorealistic Rendering (NPR), Human-Computer Interaction (HCI)
Stakeholders	Graphics researchers, 3D visualization developers
Representation Proposed	Impressionistic surface-aligned splatting (SAS). It translates raw, unstructured point cloud data into a series of "brush strokes" oriented by tangential vectors.
Communication Intention of Visual Graphics	To enhance the visual readability of sparse 3D data. By utilizing the cultural metaphor of painting, the authors intend to improve "shape perception" in situations where standard meshes or elliptical splats fail (due to noise or low resolution).
Visual Principles	Splat shape (brush stroke), color clustering (grouping), and coherent tangent vectors (orientation).
Tool/Medium	OpenGL, GLSL (for shader-based rendering), Point Cloud Library (PCL) for preprocessing.
Evaluation	Empirical comparative study (N=31, online questionnaire).
Findings on type of visualization	Brush-stroke orientation, based on coherent tangent vectors, is a superior strategy for communicating surface curvature compared to generic elliptical splats.
RQ1	The article documents critical limitations in standard point cloud rendering, specifically "background/foreground confusion" and the loss of definition in close-up or low-resolution scenarios. When sparse point clouds are rendered using traditional methods, the result is often "pixelated artifacts" that lack structural coherence. This creates a significant perceptual limitation: the eye cannot "read" the curvature or shape of the object. To address this, the authors explore the principle of Artistic Metaphorical Encoding. By shifting the paradigm from purely geometric representation to a "painterly" depiction, the principle of coherence is introduced. Through the application of the Householder formula for tangent estimation, the system ensures that individual "brush strokes" (splats) are oriented to reflect the underlying surface curvature, creating a visually continuous "flow" that guides the user's eye, thereby transforming raw noise into a readable, coherent form.
RQ2	The study leverages the precedent of Impressionist painting—an established cultural aesthetic—to substantiate its rendering choices. This is highly relevant to Embodied Carbon (EC) visualization. EC data, like raw point clouds, is often "noisy" or fragmented. The precedent here suggests that we should not aim for an "accurate" numerical visualization, but rather an "impressionistic" one. In an EC context, this means that for a building's design-stage assessment, where the data is necessarily uncertain, we could use an "impressionistic" style to communicate the overall carbon "atmosphere" of a design. If a material is carbon-intensive, the "brush stroke" might be darker or thicker; if it

	is low-carbon, it might be more refined. This precedent substantiates a shift away from high-fidelity (but often inaccurate) reporting toward expressive, stylized representations that communicate the "gestalt" or overall environmental performance of a design.
RQ3	The article discusses the limitation that traditional point-cloud visualization tools are "not suitable" for broad interaction because they require specialized knowledge to interpret artifacts or "fill in the gaps" mentally. By applying a cultural metaphor—Impressionist painting—the study broadens the potential audience. Because users are "culturally trained to interpret" this style, they can understand the geometry without needing a degree in computer graphics or point-cloud rendering. This is a crucial principle for EC communication: by using familiar artistic visual languages, we can remove the specialized technical barrier that usually surrounds LCA data. The principle of Cultural Familiarity allows the visualization to become an interface for everyone, enabling architects, clients, and even the public to engage with complex material data by relying on their innate ability to read visual "quality" rather than technical data.

Specific Approach

ID / Article Title	61 / Federated Data Modelling for Heritage Building Performance Management
Year	2026
Research type	Empirical Methodology / Applied Research (Case Study).
Research field	Heritage Building Information Modelling (HBIM), Digital Twins
Stakeholders	Building managers, conservationists
Representation Proposed	Semantic Knowledge Graphs acting as a "Federated" layer, coupled with web-based interactive dashboards.
Communication Intention of Visual Graphics	To transform disconnected, heterogeneous data (BIM geometry, sensor time-series, energy bills) into a "systemic vision" that facilitates performance-based management. The intention is to provide high-level insight (KPIs) while maintaining links to granular technical data.
Visual Principles	Dashboard-based visualisations (gauge charts, time-series line plots, 3D model viewers).
Tool/Medium	Autodesk Revit (HBIM), Neo4j (Graph Database), Python (BTwin toolkit), Plotly/Dash (web dashboards).
Evaluation	Case study demonstration (Capo di Bove complex, Rome).

Findings on type of visualization	The use of interactive dashboards that "link" 3D geometry to time-series data is the primary strategy for decision support. The study validates the transition from managing static BIM models to "living" knowledge graphs that federate multi-domain data.
RQ1	The article identifies the central limitation as "data fragmentation"—the existence of unconnected silos of information (geometry, sensor, and billing data) that prevent holistic decision-making. Standard HBIM technologies are critiqued for their inability to handle dynamic time-series data and their steep learning curve, which excludes stakeholders lacking digital expertise. To address these, the study explores the principle of Federated Data Modelling (FDM). By using Knowledge Graphs as a virtual "intermediate layer," the system creates semantic connections between disparate sources without requiring a centralized, monolithic database.
RQ2	It cites historical scientific illustration (da Vinci, Vesalius) and cartography, which balanced empirical accuracy with "ornamentation" and "symbolic decoration." Also, the influence of the Bauhaus movement in using line, color, and geometry to express the "generative aesthetic" of structural data. This suggests that EC representation can rely on using texture and form to evoke the "weight" and "impact" of specific materials. The article also points to "environmental and weather-based sculpture" (Kim et al., 2020) as a precedent for physicalizing data, offering a pathway for EC visualization to become an immersive, spatial experience rather than a digital report.
RQ3	The current graphic limitations—specifically high abstraction and "expert-oriented" complexity—exclude general audiences, thereby limiting the social impact of data. It identifies that "high abstraction may confuse general audiences", creating a barrier to participation. To democratize access, the authors propose a move toward "human-centered data storytelling" by utilizing familiar aesthetic languages—such as natural patterns, organic mappings, and intuitive symbols—the visual systems attempt to "convert ordinary communal spaces into spaces of collective thought."

ID / Article Title	67 / Visualizing ESG Performance in an Integrated GIS–BIM–IoT Platform for Strategic Urban Planning
Year	2025
Research type	Empirical Methodology / Applied Research (Case Study).
Research field	Urban Planning, Sustainability Governance (ESG), Digital Twins (DT)
Stakeholders	Urban planners, policymakers
Representation Proposed	An interactive 3D urban interface (Digital Twin). This is a geospatial representation that fuses architectural geometry with real-time sensor streams and multi-dimensional ESG data.
Communication Intention of Visual Graphics	To transform ESG assessment from a static, "indicator-heavy" reporting tool into a dynamic, "strategic asset." The intention is to make complex, siloed sustainability data visible, location-aware, and actionable for decision-makers.
Visual Principles	Geo-referenced 3D models, color-coded indicator overlays (heatmap-like thematic maps), and multi-scale navigation (zooming from city-level indices down to individual building components).

Tool/Medium	Cesium (WebGL-based 3D geospatial engine), BIM software (Revit/IFC), IoT sensor networks.
Evaluation	Case study demonstration (primary school and air quality station in Hong Kong).
Findings on type of visualization	The research confirms that semantic enrichment—mapping IoT/BIM data to a unified ontology—is the necessary prerequisite for effective urban-scale visualization. The study moves from "indicator-heavy" static reports to "interactive, location-aware" 3D interfaces, showing that decision-makers prefer holistic 3D views over granular spreadsheets.
RQ1	The article identifies a critical limitation in current ESG frameworks: they are primarily designed for corporate, non-spatial reporting, resulting in "indicator-heavy, static formats" that fail to resonate with urban planners. These formats obscure meaningful patterns and fail to integrate the spatial complexity of urban environments. Furthermore, data fragmentation is a massive barrier, as environmental, social, and governance metrics reside in different "data silos." To address these, the researchers employ the principle of Semantic Integration. By using an integrated ontology (GIS-BIM-IoT), they provide a common structural backbone that allows disparate data points to "speak the same language." The visualization principle here is multiscale contextualization—the ability to view data at the "city-scale" (governance/environmental indices) while maintaining the capability to "drill down" to the "component-scale" (building materials/sensor data). This ensures that the data is not just visualized, but structurally grounded, reducing the cognitive dissonance often caused by disconnected reporting.
RQ2	The precedent established by this paper is the Digital Twin (DT) as a "unifying visual metaphor." This is highly applicable to Life Cycle Assessment (LCA) and Embodied Carbon (EC) representation. In architecture, EC data is currently treated as an auxiliary report—separate from the design process. The precedent here suggests that EC should instead be treated as a "first-order" entity within an integrated DT environment. By mapping carbon values (IoT-like sensor stream of environmental data) onto BIM components, the EC visualization becomes an intrinsic part of the architectural entity. This substantiates a direction where EC data is visualized as a "dynamic aura" or "performance skin" draped over the building's 3D geometry. Such a strategy would allow architects to see their design choices in real-time as they relate to carbon performance, transforming the static "calculation" of carbon into an interactive "exploration" of the building's environmental footprint.
RQ3	The article discusses the limitation of "static reporting tools" that fail to attract the attention of policymakers and decision-makers who require "holistic scores." By focusing on granular, indicator-heavy data, existing systems exclude non-expert planners. The principle explored to alter this audience barrier is the "transformation of data into a strategic asset." This implies a shift in visual language from "raw numbers" (which are meaningful only to specialists) to "integrated insights" (which are meaningful to strategic governors). By using 3D urban interfaces, the system makes ESG performance "democratized"—accessible to anyone with spatial reasoning skills, rather than just those versed in complex sustainability metrics. This effectively changes the user from a "passive reader of reports" into an "active participant in urban governance." For EC communication, this principle suggests we must synthesize our data into high-level, intuitive visual indicators that clearly communicate trade-offs, enabling non-experts to influence design decisions without needing to navigate the underlying complexity of the LCA calculation.

ID / Article Title	73 / Digital Architectural Form Generation Through Pixel System-Driven Image Feature Information
Year	2024

Research type	Technical Methodology / Generative Design Research.
Research field	Computational Architecture, Generative Design
Stakeholders	Computational architects, generative design researchers
Representation Proposed	Computational Generative Modeling. It utilizes "image feature information" as a scalar or vector parameter to drive geometric form-finding in Rhino/Grasshopper.
Communication Intention of Visual Graphics	To transform 2D pixel data into 3D sculptural or architectural massing. The intention is to remove the "arbitrariness" of random form generation by establishing a causal link between image content (e.g., context photographs, historical records) and the resulting architectural volume.
Visual Principles	Voxel meshes, metaballs, Voronoi structures, and spline-based deformation.
Tool/Medium	Visual Studio Code (Python), OpenCV, Matplotlib, Microsoft Excel (as a data bridge), Rhinoceros 3D, and Grasshopper.
Evaluation	Design-research demonstration. The authors evaluate the "morphological properties" of the generated forms, assessing whether the generated forms logically follow the characteristics of the input feature data.
Findings on type of visualization	Visual strategies: Successful systematization of four categories of image information: Color, Shape, Texture, and Local Features, each producing distinct morphological outcomes (e.g., rhythmic lines for texture, random/organic point-clusters for local features).
RQ1	The article documents a failure in contemporary architectural practice: the reliance on "intuitive judgment," which the authors argue is prone to bias and leads to slow, labor-intensive design cycles. A specific limitation is that information technology in architecture is currently relegated to an "ancillary" role, used for post-design analysis rather than as a generative driver. To address these limitations, the paper explores the principle of "Information as a Core Value-Creating Tool." This principle shifts the role of data from a passive analytical output to an active generative input. By systematizing the extraction of low-level image features (color, shape, texture) and mapping these directly into a sculptural, digital design process, the study introduces a mechanism for "causal form generation." This replaces the arbitrary or purely intuitive process with a reproducible, data-driven methodology, effectively transforming raw pixel data into structural parameters.
RQ2	The article draws on precedents of generative design, where digital intermediaries connect technology to architectural outcomes. In the context of Embodied Carbon (EC) representation, this is a significant precedent. It suggests that EC is not merely a number to be reported, but a "generative driver" that could determine the physical form of a building. For instance, if carbon intensity is treated as a "texture" or "shape" feature, the framework proposed by An et al. could be adapted so that high-carbon materials force certain geometric shapes or configurations during the early design stage, effectively making the carbon footprint "morphologically inherent" in the structure. This points to a future where LCA is not a final verification step, but an generative input, much like the "pixel-feature" information in this study, which forces the designer to engage with the performance data at the earliest conceptual moment.

RQ3	The article discusses the limitation of "labor-intensive" and "intuitive-only" design processes, which essentially gatekeep high-quality design behind the expertise of experienced (and often biased) practitioners. By proposing a system that "instantly generates diverse early design alternatives," the authors alter the audience by making high-level architectural form-finding accessible to a broader range of users. The principle here is the democratization of design capability through automated, information-centric systems. This alters the intended audience by empowering early-career architects or even non-architects to explore design alternatives that would previously have required exhaustive expert manual labor. For EC communication, this suggests that if we can create similar "generative pipelines" for carbon, we could empower a wider range of stakeholders to rapidly iterate through low-carbon design alternatives, democratizing the environmental optimization process.
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ID / Article Title	93 / A Data-Driven Energy Platform: From Energy Performance Certificates to Human-Readable Knowledge through Dynamic High-Resolution Geospatial Maps
Year	2020
Research type	Empirical Methodology / Platform Development.
Research field	Data Science, Energy Informatics, Geovisualization
Stakeholders	Energy experts, public administration, and private citizens.
Representation Proposed	Geospatial, interactive data dashboards utilizing Choropleth, Scatter, and Cluster-marker maps.
Communication Intention of Visual Graphics	To resolve the "data silo" problem by federating heterogeneous EPC data into a navigable, spatial environment that supports decision-making. The intention is to translate "statistical reports" (certificates) into "perceptual maps" (city-wide performance).
Visual Principles	Geospatial positioning (spatial anchor), color-coding (energy classes/performance levels), and aggregation (cluster markers).
Tool/Medium	Python (Scikit-learn for machine learning, Folium for mapping, Flask for web application), SQLite (for database management).
Evaluation	Methodology validation via real-world case study (Piedmont Region, Italy). Algorithmic performance (accuracy of predictions/classifications), data coverage, and system usability (as defined by the ability to generate distinct, actionable insights for different stakeholder types).
Findings on type of visualization	The study confirms that geospatial clustering is essential for making large-scale building energy data legible. It validates that "coarse-grained" maps are effective for policy-level overviews, while "fine-grained" scatter maps satisfy the needs of granular analysis.
RQ1	The article documents the severe limitation of "information fragmentation," where performance data exists in isolated silos, making it inaccessible to the stakeholders who need it most. Furthermore, it critiques the failure of traditional documentation (like static

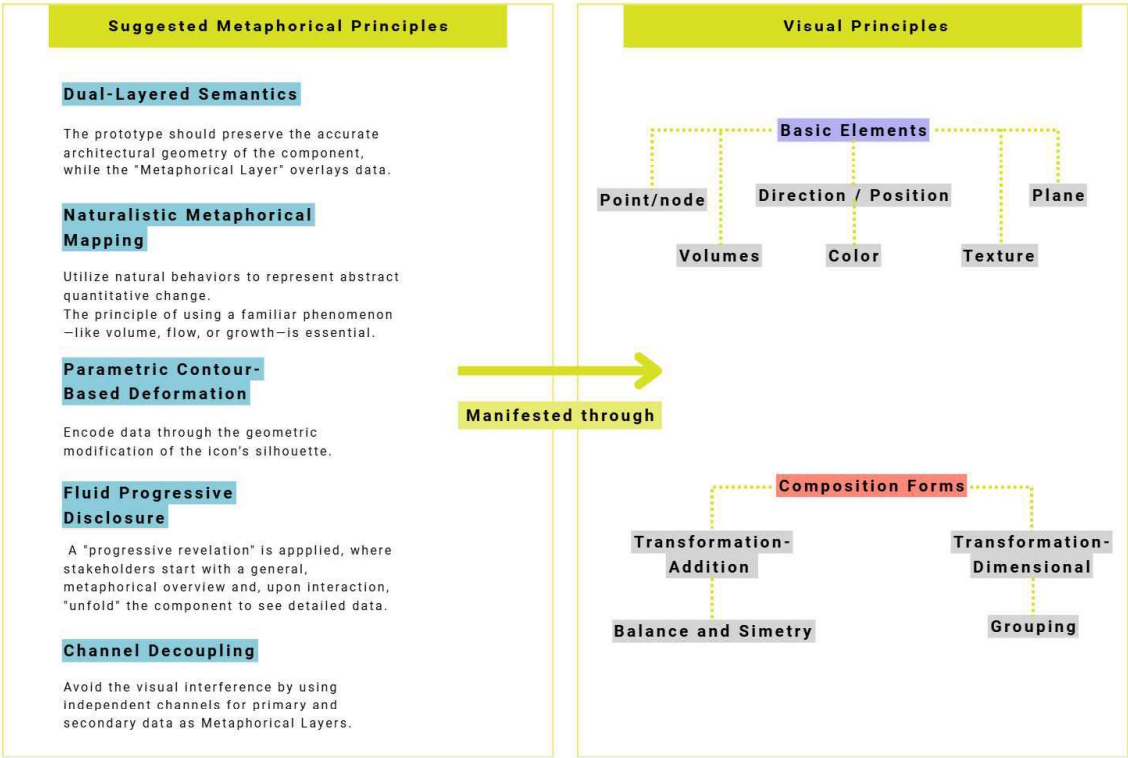
	certificates) to be useful for urban-scale strategic planning. These documents are characterized as "static" and "report-heavy," failing to reveal structural patterns or energy vulnerabilities. To address these limitations, the authors explore the principle of Geospatial Synthesis. By mapping performance metrics onto a high-resolution geospatial platform, they convert abstract numerical data into a "human-readable" visual language. This addresses the limitation of cognitive accessibility; the user is no longer required to parse complex certificates but instead interacts with a navigable map that reveals performance patterns intuitively, supporting a rapid, evidence-based decision-making process.
RQ2	The article leverages the precedent of "geospatial dashboarding," where data is embedded within the spatial context of the urban fabric. This is a critical precedent for Embodied Carbon (EC) representation. Current EC assessment is often confined to building-scale calculators, isolated from the wider urban context. The precedent of the TUCANA platform suggests that EC should be visualized at a city-district level. If material-level carbon data were geolocated and mapped as the TUCANA authors have done for energy performance, stakeholders could visualize the "carbon stock" of an entire city district. This would substantiate a move toward "carbon mapping"—a visual strategy where architectural materials are aggregated spatially, allowing planners to identify carbon-intensive zones and prioritize retrofitting or material strategies based on their geographic and structural context.
RQ3	The article discusses the limitation that "statistical representations" are often incomprehensible to non-expert end-users, despite being useful for domain specialists. To overcome this, the authors adhere to the principle of "expert-tailored presentation layers." By integrating a web application that provides specific dashboards to different user groups (citizens vs. experts), the platform effectively democratizes the data. The system limits the visual complexity for the general public (using intuitive maps) while providing the technical "hooks" for domain experts. This is a crucial principle for EC communication: we cannot expect a single visualization to serve both the developer and the citizen. We must adopt a multi-tier approach, where the "Human-Readable Knowledge" (the target) is prioritized for non-experts, while the raw technical data remains accessible but visually subordinated in the system's architecture.

9.3 Data Extraction Analysis of AI Studio

Following the completion of the data extraction phase and the submission of the articles to AI-based scrutiny, alternative visual strategies for complex data visualization were identified with the assistance of AI Studio, particularly in light of the cognitive limitations inherent in conventional methods.

We refer to these identified strategies as 'Metaphorical Principles,' from which we selected those that collectively underpin our MET-VIS proposal. Furthermore, we identified the corresponding 'Visual Principles' that manifest each Metaphorical Principle.

This relationship is illustrated in the figure below:



For enhanced contextualization, the following tables indicate the source articles from which each Metaphorical Principle and Visual Principle was derived:

Metaphorical Principles

Metaphorical Principles	(ID) Article related
Dual-Layered Semantics	(03) DATA VISUALIZATION AS ARTISTIC EXPRESSION (61) Federated Data Modelling for Heritage Building Performance Management (67) Visualizing ESG Performance in an Integrated GIS–BIM–IoT Platform for Strategic Urban Planning

Naturalistic Metaphorical Mapping	(12) Visualization Metaphors in the Tasks of Exploratory Analysis of Heterogeneous Data
Parametric Contour-Based Deformation	(11) Pacemod: parametric contour-based modifications for glyph generation
Fluid Progressive Disclosure	(05) Fluidly Revealing Information: A Survey of Un/foldable Data Visualizations
Channel Decoupling	(16) Visualizing ordered bivariate data on node-link diagrams

Visual Principles

Visual Principles	(ID) Article related
Point / Node	(08) Improve Usability of Multidimensional Data Exploration with Water Fountain Based 3D User Interface Metaphor (09) Designing complex network visualisations using the wayfinding map metaphor (10) Visually encoding orthogonal planar graph drawings as graph mazes: An eye tracking study (16) Visualizing ordered bivariate data on node-link diagrams
Direction / Position	(09) Designing complex network visualisations using the wayfinding map metaphor (10) Visually encoding orthogonal planar graph drawings as graph mazes: An eye tracking study
Plane	(12) Visualization Metaphors in the Tasks of Exploratory Analysis of Heterogeneous Data
Volume	(03) DATA VISUALIZATION AS ARTISTIC EXPRESSION (05) Fluidly Revealing Information: A Survey of Un/foldable Data Visualizations (08) Improve Usability of Multidimensional Data Exploration with Water Fountain Based 3D User Interface Metaphor (10) Visually encoding orthogonal planar graph drawings as graph mazes: An eye tracking study

	(12) Visualization Metaphors in the Tasks of Exploratory Analysis of Heterogeneous Data
Color	(03) DATA VISUALIZATION AS ARTISTIC EXPRESSION (08) Improve Usability of Multidimensional Data Exploration with Water Fountain Based 3D User Interface Metaphor (11) Pacemod: parametric contour-based modifications for glyph generation (16) Visualizing ordered bivariate data on node-link diagrams
Texture	(03) DATA VISUALIZATION AS ARTISTIC EXPRESSION (11) Pacemod: parametric contour-based modifications for glyph generation (16) Visualizing ordered bivariate data on node-link diagrams
Transformation - Addition	(03) DATA VISUALIZATION AS ARTISTIC EXPRESSION
Transformation - Dimensional	(05) Fluidly Revealing Information: A Survey of Un/foldable Data Visualizations (08) Improve Usability of Multidimensional Data Exploration with Water Fountain Based 3D User Interface Metaphor (11) Pacemod: parametric contour-based modifications for glyph generation
Balance and Simetry	(03) DATA VISUALIZATION AS ARTISTIC EXPRESSION
Grouping	(05) Fluidly Revealing Information: A Survey of Un/foldable Data Visualizations (08) Improve Usability of Multidimensional Data Exploration with Water Fountain Based 3D User Interface Metaphor (09) Designing complex network visualisations using the wayfinding map metaphor

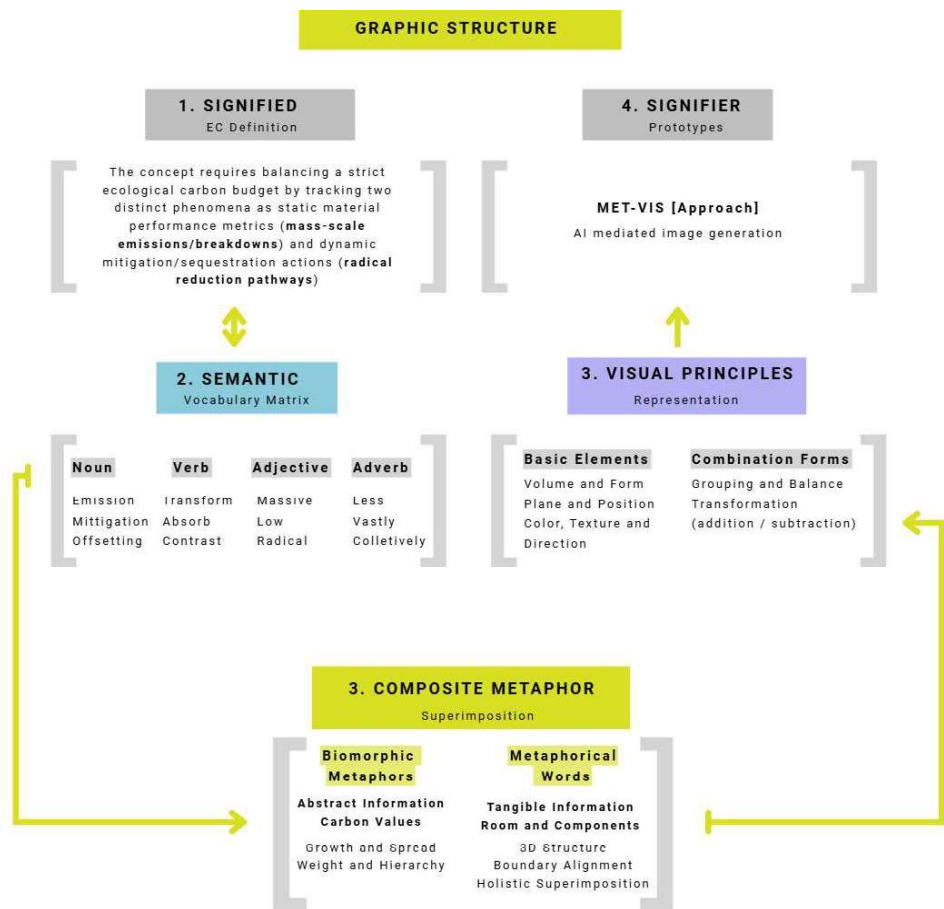
9.4 Proposed Graphic Structure

Finally, from the SLR and incorporating AI systematization, we imagined the abstract concepts that existed related to EC that could be transported into tangible graphic rules.

Beginning with a generic EC definition, the signified of the graphic structure, we identified the main semantic attributes to explain this indicator. The Vocabulary Matrix partitions EC into tangible, text-based directions, solidifying abstract concepts into more approachable ideas. For this purpose, we considered: scope, solution (nouns), urgent or strategic actions (verbs), qualitative, scale, impact (adjectives), and temporal, degree, and comparative (adverbs).

Then a Composite Metaphor emerged from connecting the Vocabulary Matrix and the SLR-DE findings, encompassing two superimposed metaphor types: Biomorphic Metaphor for carbon data and Metaphorical Worlds for architectural setup. Derived Visual Principles provided a graphic-based, tangible language best suited to the metaphors, finally arriving at the signifier of the graphic structure: MET-VIS approach based on AI-mediated exploratory images.

The figure below depicts the process synthesis for our experimental MET-VIS perspective:



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